Case Study Description: FinTrust Digital Financial Services Platform

Background

FinTrust is a fictional fintech company created for this database project. It provides digital financial services that allow customers to manage accounts and carry out transactions such as deposits, withdrawals and transfers.

As the number of customers and transactions increases, the company needs a database that can organise its records properly and make information easy to retrieve.

Problem Statement

Managing customer details, account information and transaction records without a properly structured database can lead to duplicate records, inconsistent information and difficulties tracking financial activities.

FinTrust therefore requires a relational database to keep these records organised and connected.

Aim and Objectives

The aim of this project is to design and implement a relational database for managing FinTrust's customer and financial records.

The database is designed to:

- Store customer information.
- Maintain customer account details and balances.
- Record deposits, withdrawals and transfers.
- Track transaction types and statuses.
- Keep records of important activities through audit logs.
- Maintain relationships between related records and reduce data duplication.

Scope of the Database

The database covers six main areas:

- **Customers:** Stores customer details such as names, email addresses and phone numbers.
- **Accounts:** Stores account numbers, account types, balances and account status.
- **Transaction Types:** Identifies deposits, withdrawals and transfers.
- **Transaction Statuses:** Tracks whether transactions are pending, successful or failed.
- **Transactions:** Stores transaction references, amounts, dates and related account information.
- **Audit Logs:** Records important activities carried out within the system.

The project focuses on the database design and implementation using MySQL. It does not cover the development of a complete banking application or connection to a real payment system.

Tools Used

- MySQL 8.0
- MySQL Workbench
- Git and GitHub

Note: FinTrust is a fictional organisation used strictly for academic purposes. All sample customer and transaction records were created for this project.